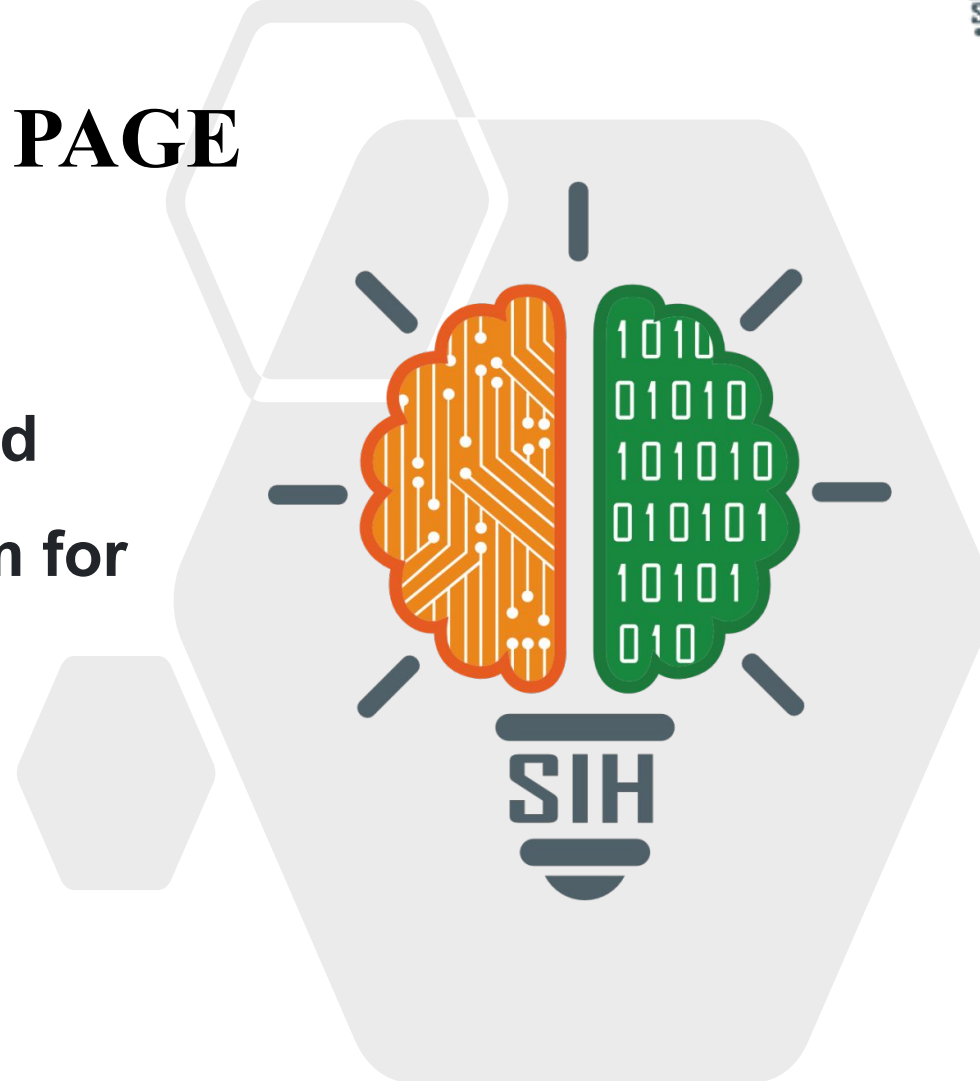


SMART INDIA HACKATHON 2025



TITLE PAGE

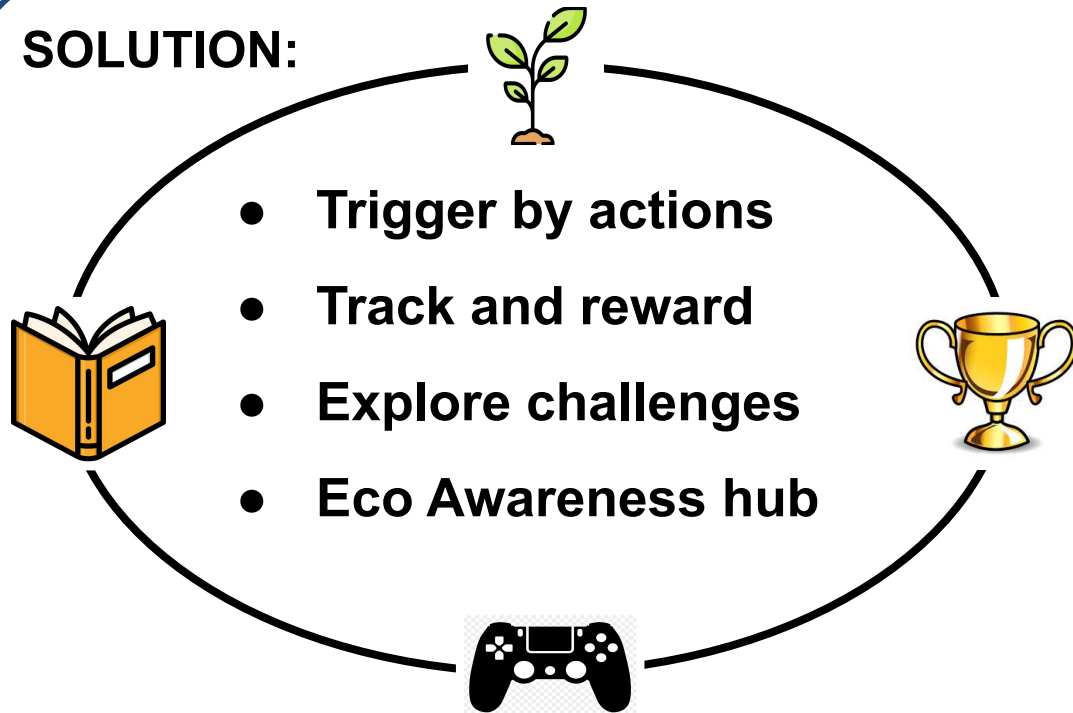
- **Problem Statement ID – 25009**
- **Problem Statement Title - Gamified Environmental Education Platform for Schools and Colleges**
- **Theme - Smart Education**
- **PS Category- Software**
- **Team ID -**
- **Team Name-New Tech Titans**



PROBLEM:

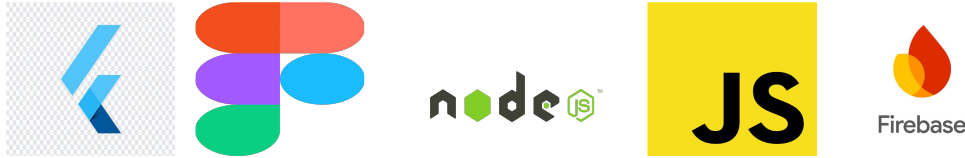
Environmental education in schools and colleges is largely theoretical and unengaging, leading to low student interest, poor awareness, and minimal adoption of sustainable habits needed to address urgent ecological challenges

SOLUTION:



INNOVATION:

- **Real world integration** - Rewards linked to real eco-actions
- **Adaptive learning** - Personalized tasks based on age group and curriculum
- **Collaborative missions** - Team challenges across schools/colleges to build eco-communities
- **Green impact tracker** - Visualizes eco-impact, carbon saved, waste cut, trees planted.



FRONT END:

- **Framework:**Flutter(cross-platform mobile app)
- **Language:**Dart
- **Design tools:**Figma(for UI/UX prototyping)

BACKEND:

- **Framework:**Node.js
- **Language:**Javascript
- **Database:**Firebase realtime DB

GAMIFICATION ENGINE:

- Points,badges,leaderboard system
- Progress tracking and reward algorithm
- AI-based personalized learning paths



MOBILE SENSORS/INTEGRATION:

- **Camera:**For uploading proof
- **GPS:**For location and activity tracking

APIs AND SERVICES:

- **Google maps API:**Location-based activities and eco-mapping
- **cloud hosting:**AWS/Azure/GCP for scalability and reliability

SECURITY AND AUTHENTICATION:

- **Role based access:** For students,teachers,admin
- **OAuth 2.0/JWT:** For secure login



TECHNICAL FEASIBILITY

CLOUD+MOBILE/WEB
PLATFORM SCALABLE,
GAMIFICATION READY



OPERATIONAL FEASIBILITY

EASY ADOPTION IN
SCHOOLS-TEACHER
DASHBOARD+REPORT



ECONOMIC FEASIBILITY

LOW COST CLOUD
HOSTING AFFORDABLE
FOR INSTITUTIONS

CHALLENGES:



- DIGITAL DIVIDE
- INSTITUTIONAL RESISTANCE
- ENGAGEMENT RISK
- DATA PRIVACY AND SECURITY



STRATEGIES:



- BALANCED ENGAGEMENT



- PARTNERSHIP



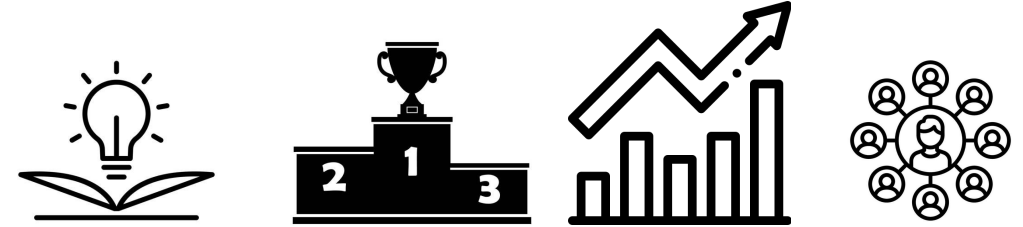
- ACCESSIBILITY



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IMPACTS:

- **Enhanced Environmental Awareness:** practical knowledge eco-friendly practices.
- **Behavioral Change:** Encourages adoption of green habits.
- **Collaborative Learning:** Promotes teamwork through eco-challenges.
- **Youth Empowerment:** Cultivates future eco-leaders.



BENEFITS:

- **Engaging Learning:** Make environmental education fun and interactive.
- **Recognition & Motivation:** Badges, points, and leaderboards motivate continued participation.
- **Institutional Advantage:** Track student progress and showcase sustainability initiatives.
- **Scalability & Reach:** Making environmental education accessible to a wider audience.

RESEARCH:

- **Byju's:** Uses interactive modules to boost student engagement in learning.
- **Duolingo:** Gamified app with streaks, levels, and rewards for habit-building.
- **EcoHero:** Rewards eco-friendly challenges like recycling and saving energy.
- **WWF Free Rivers (AR):** Teaches ecosystems through immersive AR experiences.
- **JouleBug:** Makes sustainable living fun by turning daily eco-friendly actions into shareable challenges.

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